









Modelling Financial Markets

Chapter 4: Portfolio Optimisation



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Why Does Estimating the Optimum Change the Problem?

Easy optimum, hard estimate

- A **portfolio** is a weight vector $\mathrm{w} = (w_1, \ldots, w_N)^\top$ with $\mathrm{w}^\top 1 = 1$; optimisation chooses w to maximise a criterion under constraints
- The optimum is easy to write with known means and covariances, hard to estimate: the optimiser loads on the largest estimation errors
  - 16 assets: 16 means + 136 variances and covariances = 152 parameters, estimated from 60 monthly observations
- Can any estimated rule beat equal weights out of sample, after costs?

How Will We Compare Portfolio Rules?

What we will explain

- Part I: mean–variance theory, estimation risk, covariance shrinkage, Black–Litterman
- Part II: risk parity and Hierarchical Risk Parity (HRP), out-of-sample tests, portfolios on the Bucharest Stock Exchange (BVB), a case study, the research loop assisted by artificial intelligence (AI)
- Technical detail: Appendix; textbook: Franke, Härdle & Hafner (2019), Part I

What you should be able to do

- compute minimum-variance, tangency and equal-risk weights
- explain why estimation error favours 1/N
- test a Sharpe difference after costs and multiple testing

Two Routes Through the Chapter

Core path

- Mean–variance: GMV and tangency weights, the frontier, constraints
- Estimation risk: why in-sample Sharpe ratios are too optimistic
- Three repairs of the inputs: covariance shrinkage, constraints, Black–Litterman
- Rules that need only Σ: ERC and HRP
- Out-of-sample evaluation: turnover and costs, Sharpe-difference tests, Holm and Benjamini–Hochberg; then the BVB application

Deeper study

- The case study on trading costs for a large investor, and the AI research loop with its data-snooping mini-case
- Appendix: derivations (frontier, Wishart moments, Britten-Jones), the complete result tables and all 21 tests against 1/N
- Quick checks with toy numbers close the chapter; answers in the Appendix

Data, Assets and the 1/N Benchmark

Universe	Assets	Returns	Currency
Sectors	9 US sector ETFs	1999/01 – 2026/07	USD
Multi-asset	8 ETFs: equities (3), Treasuries (2), credit (2), gold	2007/05 – 2026/07	USD
Combined	9 sectors + the 7 multi-asset ETFs other than SPY (S&P 500) (16 ETFs)	2007/05 – 2026/07	USD
Romania	8 BVB blue chips; the BET and BET-TR indices	2014/10 – 2026/08	RON

- ETF (exchange-traded fund): a fund traded on an exchange that tracks an index
- Monthly simple returns from month-end prices on common trading days
  - ETFs and stocks with dividends reinvested
  - BET (Bucharest Exchange Trading), the BVB benchmark, is a price index; BET-TR (total return) reinvests dividends
- Excess returns over the one-month Treasury bill (Kenneth French Data Library); BVB: $R_f = 0$
- **1/N**: equal weights $w_i = 1/N$, rebalanced every month; no estimation at all, the benchmark of every comparison

The Assets in Detail

- **Sectors** (9 US sector ETFs)
  - XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology
  - XLP staples, XLU utilities, XLV health care, XLY discretionary
- **Multi-asset**
  - equities: SPY (S&P 500), EFA (developed markets outside the US and Canada), EEM (emerging markets)
  - bonds: TLT (US Treasuries, 20+ years), IEF (US Treasuries, 7–10 years), LQD (investment grade), HYG (high yield); gold: GLD
- **BVB** (8 blue chips)
  - TLV Banca Transilvania, SNP OMV Petrom, BRD BRD–Groupe Société Générale, TGN Transgaz
  - SNG Romgaz, SNN Nuclearelectrica, EL Electrica, TEL Transelectrica

Notation (1/2)

- **Returns**
  - simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; $P_{i,t}$: the month-end price of asset i
  - excess return $R^e_{i,t} = R_{i,t} - R_{f,t}$ (in the seminar: r_t and r^e_t); $R_{f,t}$: the risk-free rate
  - log return $\ln(P_{i,t}/P_{i,t-1}) = \ln(1 + R_{i,t})$: adds up over time, not across assets; e.g. a price from 100 to 110: $R = 10\%$, $\ln 1.1 = 9.53\%$
  - portfolios use simple returns, because $R_{p,t} = \sum_i w_i R_{i,t}$ holds exactly
- **Portfolio** of N assets
  - w: vector of weights; $\boldsymbol{\mu}$: vector of expected (excess) returns; 1: vector of ones
  - $\boldsymbol{\Sigma}$: the $N \times N$ covariance matrix of the returns
  - portfolio mean $\mathrm{w}^\top \boldsymbol{\mu}$, portfolio variance $\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$

Notation (2/2)

- **Sharpe ratio** (Sharpe (1966)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$
  - the mean excess return per unit of volatility σ
  - monthly figures annualised with $\times 12$ (mean) and $\times\sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t(W_t/\max_{s\le t} W_s - 1)$
  - the largest fall of wealth W_t from a previous peak
  - example: wealth rises to 1.50, then falls to 1.20: $1.20/1.50 - 1 = -20\%$
- **Out of sample** (OOS): the weights held in month t use only data up to month $t-1$
- **Rules of this chapter**
  - MV (mean–variance): tangency portfolio from sample moments; GMV: global minimum variance
  - MV-LO and GMV-LO: their long-only (LO) versions; GMV-LW: GMV with Ledoit–Wolf (LW) covariance shrinkage
  - ERC (equal risk contribution) and HRP: risk-based rules; 1/N: equal weights

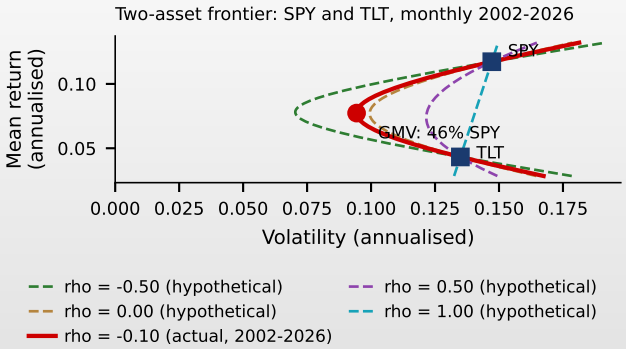
The Markowitz Problem

- Markowitz (1952): investors care about the mean and the variance of portfolio returns
- **Minimum-variance frontier**: minimum variance for every target mean m
  - its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_{\mathrm{w}} \ \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} \quad \text{s.t.} \quad \mathrm{w}^\top \boldsymbol{\mu} = m, \ \ \mathrm{w}^\top 1 = 1$$

  - Objective: portfolio variance; constraints: the target mean m and full investment (weights sum to 1)
- Utility form, risk aversion $\gamma > 0$: $\max_{\mathrm{w}} \ \mathrm{w}^\top \boldsymbol{\mu} - \frac{\gamma}{2} \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top 1 = 1$
  - as γ varies, the solution traces exactly the efficient branch
- Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: How Correlation Diversifies



- Two assets, correlation ρ: $\sigma_p^2 = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$; w: weight of asset 1, σ_1, σ_2: volatilities, σ_p: portfolio volatility
- Minimised at $w^* = \dfrac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026: $\rho = -0.10$, $w^* = 46\%$ SPY, volatility 9.4% against SPY 14.7% and TLT 13.5%
- Chart: means of total returns and annualised volatilities, in decimals (0.10 = 10%)
- The lower ρ, the more the curve bends to the left; at $\rho = 1$ the long-only portfolios shown have no diversification benefit
- Time-varying correlation: Chapter 6

MFM_ch4_two_asset

GMV and Tangency Portfolios for N Assets

- GMV: $\min_{\mathrm{w}} \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top 1 = 1$; Lagrangian $\mathcal{L} = \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} - \lambda(\mathrm{w}^\top 1 - 1)$, $\boldsymbol{\Sigma}$ positive definite
  - λ: Lagrange multiplier of the budget constraint; first-order condition $2\boldsymbol{\Sigma}\mathrm{w} = \lambda 1$, so $\mathrm{w} = \frac{\lambda}{2}\boldsymbol{\Sigma}^{-1}1$; the constraint gives $\frac{\lambda}{2} = 1/A$, $A = 1^\top \boldsymbol{\Sigma}^{-1} 1$
- Tangency: maximise $\mathrm{w}^\top \boldsymbol{\mu} / \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}$, $\boldsymbol{\mu}$ the expected **excess** returns; the condition $\boldsymbol{\mu} \propto \boldsymbol{\Sigma}\mathrm{w}$ gives $\mathrm{w} \propto \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ ($\propto$: proportional to); θ: the maximum Sharpe ratio

	GMV	**Tangency (MV)**
Weights	$\mathrm{w}_{\mathrm{GMV}} = \boldsymbol{\Sigma}^{-1}1/A$	$\mathrm{w}_{\mathrm{MV}} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/B$, $B = 1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$
Result	variance $\sigma^2_{\mathrm{GMV}} = 1/A$	squared Sharpe $\theta^2 = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$
Needs	$\boldsymbol{\Sigma}$ only	$\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$; Sharpe $\mathrm{sign}(B)\,\theta$

- The normalised $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/B$ is the maximum only if $B > 0$; $B < 0$ gives the minimum-Sharpe portfolio, $B = 0$ no fully invested solution

Separation and the Frontier in Closed Form



James Tobin, Nobel Prize in Economics 1981

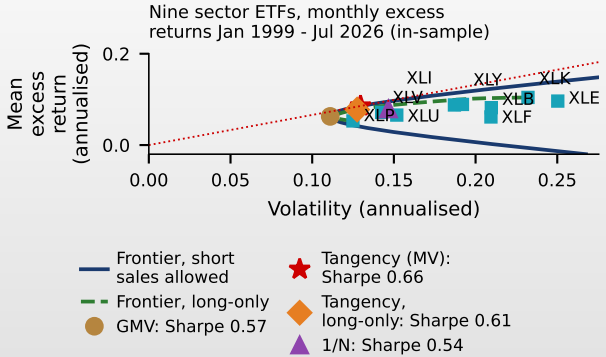
Photo: unknown author (1962); public domain; Wikimedia Commons

- Tobin (1958): add a riskless asset paying R_f
  - every investor holds the **same** risky portfolio, the tangency portfolio, mixed with cash or borrowing
  - risk aversion changes only the mix: **two-fund separation**, the bridge to the Capital Asset Pricing Model (CAPM; Chapter 3)
- Merton (1972): with short sales, the frontier is a hyperbola; Σ positive definite, $\boldsymbol{\mu}$ not proportional to 1
  - A, B as on the previous slide; $C = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = \theta^2$, $D = AC - B^2 > 0$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- $\sigma^2(m)$: the smallest variance attainable with mean m; a parabola in the (σ^2, m) plane, the hyperbola above in the (σ, m) plane (derivation: ▸ Appendix)
  - the GMV sits at $m = B/A$, $\sigma^2 = 1/A$
  - every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency (if $B \neq 0$)

The Frontier of US Sectors



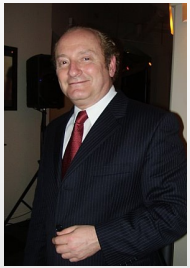
- Monthly excess returns, January 1999 – July 2026 ($T = 331$), axes annualised, in decimals; all parameters estimated on the full sample: an **in-sample** picture
- Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); 1/N: 0.54
- Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care: short positions
- Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero, a **corner solution**
- The gain over 1/N is small even with perfect hindsight

MFM_ch4_frontier

Constraints

- **Long-only** (LO): $w_i \geq 0$; no closed form, solved by quadratic programming
- Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$, G a group of assets, c_G its cap
- Leverage limit: $\sum_i |w_i| \leq L$; $\sum_i |w_i|$: gross exposure, $L = 1$ means no short sales
- Frost & Savarino (1988): constraining weights improved out-of-sample performance; why this works is a shrinkage result (next section)

Means Are Hard to Estimate: Merton's Argument

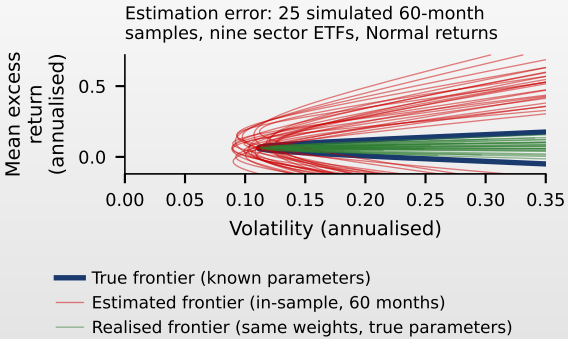


Robert C. Merton, Nobel Prize in Economics 1997

Photo: Digarnick (2006); public domain; Wikimedia Commons

- Merton (1980): Y years sampled h times a year, i.i.d. Normal returns (a diffusion) with annual mean μ and variance σ^2
  - $\mathrm{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h; only the **span** of the sample helps
  - $\mathrm{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \to 0$ as $h \to \infty$ (Normal; in general $(\kappa - 1)\sigma^4/(hY)$, κ the kurtosis): variances sharpen with frequency
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%, 95% interval $[5.8\%, 17.6\%]$ for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

Error Maximisation: Estimated versus Realised Frontiers



- Michaud (1989): optimisers are **estimation-error maximisers**; Best & Grauer (1991): a tiny change in one mean moves many weights
- True parameters = sector moments 1999–2026; draw 60 Normal months, optimise, evaluate with the true parameters
- Chart: 25 simulated 60-month samples; red: estimated frontiers, too optimistic; green: the same portfolios under the true parameters, all below the true frontier
- 2000 simulated tangency portfolios: median estimated maximum Sharpe 1.56, median true Sharpe 0.30, true optimum 0.66
- 96% have a true Sharpe below 1/N (0.54), 7.5% below zero; in 11% of samples $1^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}} < 0$

MFM_ch4_estimation_error

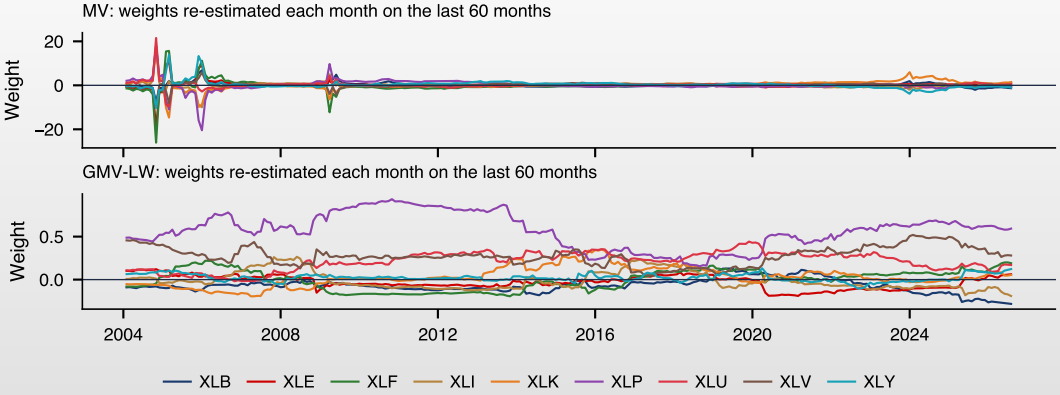
The Plug-in Sharpe Ratio Is Biased

- i.i.d. $N(\mu, \Sigma)$ returns, T months, N assets (Kan & Zhou (2007); Wishart moments in the Appendix)
  - $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$: the in-sample squared maximum Sharpe ratio, from the maximum-likelihood $\hat{\mu}$, $\hat{\Sigma}$; θ^2: its true value

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T-N-2}\left(\theta^2 + \frac{N}{T}\right) = \frac{T\theta^2 + N}{T-N-2}, \qquad T > N+2$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\sqrt{12\,\mathbb{E}[\hat{\theta}^2]} = 1.65$ a year, a root mean square, not a median
  - the simulated median 1.56 is lower: $\hat{\theta}$ is right-skewed, and the simulation uses the divisor $T-1$ in $\hat{\Sigma}$; Seminar 4, A8 derives the formula, and 1.65 is its check value
- Unbiased estimator of θ^2: $\hat{\theta}^2_u = \left[(T-N-2)\,\hat{\theta}^2 - N\right]/T$
  - It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and $\sqrt{\hat{\theta}^2_u}$ is not unbiased for θ
- Sectors, full sample ($T = 331$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}^2_u} = 0.31$ a year
  - more than half of the in-sample Sharpe is noise, even with 27 years
- The tangency weights are the ordinary least squares (OLS) coefficients of a regression of 1 on the returns (Britten-Jones (1999); Appendix; Seminar 4, A9 and B6)

Unstable Weights (1/2)



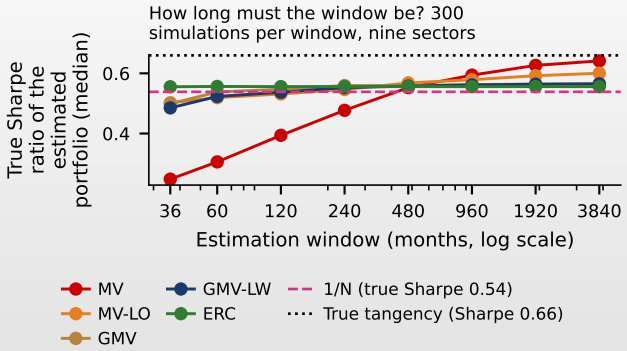
MFM_ch4_oos_backtest

Unstable Weights (2/2)

- Nine sector ETFs; each month the weights use only the previous 60 months; top panel clipped at ± 6
- MV: median gross exposure $\sum_i |w_i| = 4.9$, above 3 in 90% of the months; largest single weight 26 in absolute value
- Explosions in 2004–2006 and 2009: the denominator $1^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$ is close to zero, so the tangency weights have very heavy tails
  - Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- GMV-LW (bottom): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

MFM_ch4_oos_backtest

How Long Must the Window Be to Beat 1/N?



- DeMiguel, Garlappi & Uppal (2009): 14 models on 7 data sets; none beats 1/N consistently in Sharpe ratio, certainty-equivalent return (mean $-\frac{\gamma}{2}\times$ variance, the riskless return worth the same utility) or turnover; their calibration needs about 3000 months for 25 assets
- Here: 300 simulated samples per window, population parameters = sector moments
- MV beats 1/N only with 480 months (40 years): median true Sharpe 0.55 vs 0.54; with 60 months 0.31
- MV-LO, GMV, GMV-LW are close to 1/N from 60 months on; ERC is almost insensitive, 0.56
- In this calibration 1/N is nearly efficient (true tangency 0.66): simulation evidence, not an empirical test

MFM_ch4_window_length

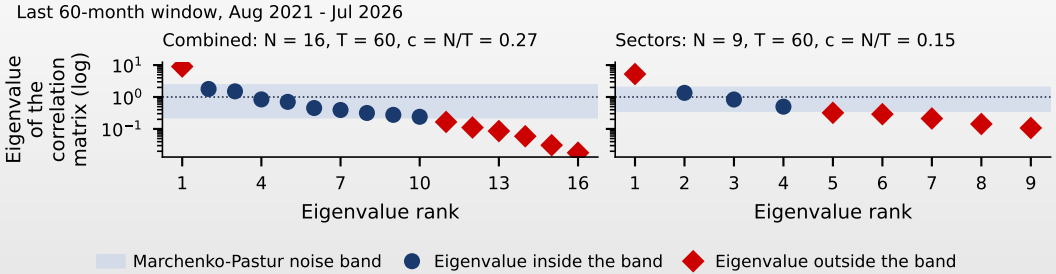
Why the Sample Covariance Matrix Fails

- Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- **Condition number**: largest over smallest eigenvalue; a large value means $\boldsymbol{\Sigma}^{-1}$ amplifies noise
- Optimisers invert $\boldsymbol{\Sigma}$, so they load on the directions with the smallest estimated variance
- The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$
- **Marchenko–Pastur** benchmark (Marčenko & Pastur (1967)): i.i.d. returns with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ (I: identity matrix)
  - as $N, T \to \infty$ with $N/T \to c \in (0, 1)$, the sample eigenvalues fill

$$[\lambda_-, \lambda_+] = \left[\sigma^2(1 - \sqrt{c})^2,\ \sigma^2(1 + \sqrt{c})^2\right]$$

  - λ_-, λ_+: the smallest and largest eigenvalue that pure noise produces
  - $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
  - Laloux et al. (1999): eigenvalues outside the band are candidates for structure; those inside cannot be told from noise by this comparison alone

Sample Eigenvalues versus the Noise Band



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); shaded band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$
- Only the market mode lies above λ_+: 9.0 (56% of the trace) for 16 ETFs, 5.2 for nine sectors ($\lambda_+ = 1.92$); with the noise variance rescaled to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)), three eigenvalues exceed the band in each universe
- Six of 16 eigenvalues lie **below** the band: near-riskless combinations such as TLT against IEF; GMV loads exactly on them, so if they are underestimated, so is its risk

MFM_ch4_inference

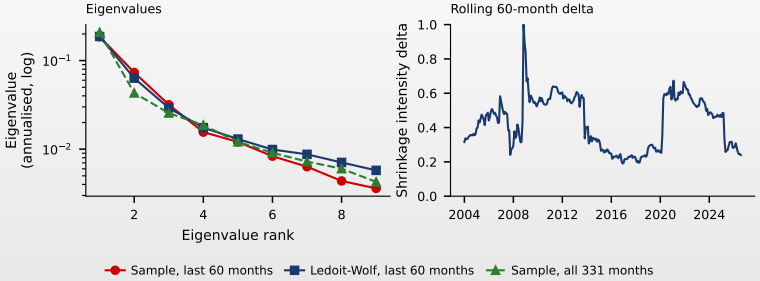
The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\Sigma}_{\mathrm{LW}} = \delta\,\mathrm{F} + (1-\delta)\,\mathrm{S}, \qquad 0 \le \delta \le 1$$

- Targets
  - Single-index (market) model: Ledoit & Wolf (2003)
  - Scaled identity: Ledoit & Wolf (2004a)
  - Constant correlation, $f_{ii} = s_{ii}$ and $f_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}}$ for $i \neq j$, $\bar{r}$ the average sample correlation: Ledoit & Wolf (2004b), used here
- δ trades the variance of S against the bias of F: noisy data or a good target, large δ; long samples, $\delta \to 0$ (the optimal δ^* in the Appendix)

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged; condition number 51.6 $\rightarrow$ 32.5
- Rolling δ: mean 0.43, range 0.19–1.00; $\delta = 1$ in the early windows, where 60 months are too few to beat the target
- GMV-LW: the GMV portfolio computed with $\hat{\boldsymbol{\Sigma}}_{\mathbf{LW}}$

MFM_ch4_shrinkage

Constraints Are Shrinkage Too

- Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\boldsymbol{\lambda} 1^\top + 1\boldsymbol{\lambda}^\top), \qquad \mathcal{L} = \tfrac{1}{2} w^\top S w - \lambda_0 (w^\top 1 - 1) - \boldsymbol{\lambda}^\top w$$

  - $\boldsymbol{\lambda} \geq 0$: the Lagrange multipliers of the constraints $w_i \geq 0$ in $\mathcal{L}$; $\lambda_i > 0$ only where the constraint binds
  - The first-order condition $Sw = \lambda_0 1 + \boldsymbol{\lambda}$ and $\boldsymbol{\lambda}^\top w = 0$ give $\tilde{S}w = \lambda_0 1$: the GMV condition for $\tilde{S}$
  - step by step, with the budget $1^\top w = 1$: $\tilde{S}w = Sw - \boldsymbol{\lambda}(1^\top w) - 1(\boldsymbol{\lambda}^\top w) = \lambda_0 1 + \boldsymbol{\lambda} - \boldsymbol{\lambda} - 0$
- $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers its covariances, which shrinks the large estimates
  - why there: the unconstrained GMV shorts assets with high covariances with the others, so $w_i \geq 0$ binds on exactly those assets, whose large covariance estimates are the most likely to be inflated by noise
- Imposing the "wrong" constraint (no short sales) can reduce out-of-sample risk; the effect is largest when covariances come from few observations

Constraints as Shrinkage: A Two-Asset Example

- $\sigma_1 = 25\%$, $\sigma_2 = 10\%$, $\rho = 0.5$: $s_{11} = 0.0625$, $s_{22} = 0.0100$, $s_{12} = 0.0125$
- Unconstrained GMV: $w_1^* = (s_{22} - s_{12})/(s_{11} + s_{22} - 2s_{12}) = -0.0025/0.0475 = -0.053$
  - the volatile asset is shorted because it co-moves strongly with the other; volatility 9.93%
- Long-only: $w_1 \geq 0$ binds, w $= (0, 1)$, volatility 10%
  - first-order condition: Sw $= (s_{12}, s_{22}) = (0.0125, 0.0100) = \lambda_0 1 + \boldsymbol{\lambda}$, so $\lambda_0 = 0.0100$, $\boldsymbol{\lambda} = (0.0025, 0)$
- Modified matrix: $\tilde{s}_{12} = 0.0125 - 0.0025 = 0.0100$, $\tilde{s}_{11} = 0.0625 - 2(0.0025) = 0.0575$, $\tilde{s}_{22} = 0.0100$
  - correlation lowered from 0.5 to $0.0100/\sqrt{0.0575 \times 0.0100} \approx 0.42$; the variance of asset 1 also falls
  - unconstrained GMV of $\tilde{\mathrm{S}}$: $w_1^* = (0.0100 - 0.0100)/(\cdots) = 0$, the long-only solution

Nonlinear Shrinkage

- Linear shrinkage mixes two matrices with one δ; **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
  - Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Out of sample, 60-month window, GMV volatility
  - Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
  - 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix
- A constant correlation fits neither stock–stock (positive) nor stock–bond (negative) pairs: the target matters (Seminar 4, B7, which uses the six volatilities above as its replication target)
- Rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

From Covariance Shrinkage to Mean Shrinkage



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Shrinkage fixes $\hat{\boldsymbol{\Sigma}}$; Black & Litterman (1992), developed at Goldman Sachs, shrinks the means towards those implied by a **benchmark** portfolio w_b
- Reverse optimisation: a mean–variance investor with risk aversion $\delta_{\mathbf{RA}}$ and access to cash holds $w^* = (\delta_{\mathbf{RA}} \boldsymbol{\Sigma})^{-1} \boldsymbol{\mu}$ in risky assets; the returns that make w_b optimal are

$$\boldsymbol{\pi} = \delta_{\mathbf{RA}} \boldsymbol{\Sigma} w_b$$

- $\boldsymbol{\pi}$: **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{\mathbf{RA}} = 2.5$
- Then add the investor's **views**, weighted by their confidence: the weights move away from the benchmark only where views exist

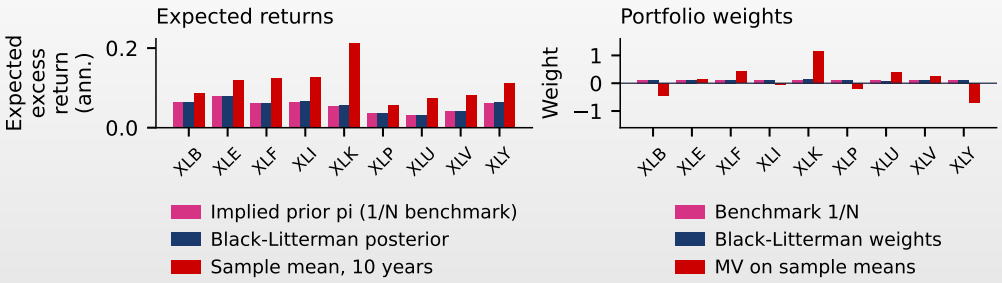
Views and the Posterior

- K views: $\mathrm{P}\boldsymbol{\mu} = \mathrm{q} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(0, \boldsymbol{\Omega})$; prior $\boldsymbol{\mu} \sim N(\boldsymbol{\pi}, \tau\boldsymbol{\Sigma})$
  - P: $K \times N$ matrix, row k = the portfolio of view k; q: the view returns; $\boldsymbol{\Omega}$: their uncertainty; τ: scales the uncertainty of the prior $\boldsymbol{\pi}$
- Posterior mean: a precision-weighted average; weights $\mathrm{w}_{\mathbf{BL}} = (\delta_{\mathbf{RA}}\boldsymbol{\Sigma})^{-1}\boldsymbol{\mu}_{\mathbf{BL}}$

$$\boldsymbol{\mu}_{\mathbf{BL}} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{P}\right]^{-1}\left[(\tau\boldsymbol{\Sigma})^{-1}\boldsymbol{\pi} + \mathrm{P}^{\top}\boldsymbol{\Omega}^{-1}\mathrm{q}\right]$$

- Limits: $\boldsymbol{\Omega} \to 0$ gives $\mathrm{P}\boldsymbol{\mu}_{\mathbf{BL}} = \mathrm{q}$ exactly; $\boldsymbol{\Omega} \to \infty$ gives $\boldsymbol{\pi}$; without views, the benchmark itself
- He & Litterman (2002): $\tau = 0.05$, $\boldsymbol{\Omega} = \mathrm{diag}(\mathrm{P}\,\tau\boldsymbol{\Sigma}\,\mathrm{P}^{\top})$; with $\boldsymbol{\Sigma}$ in the weights, only the assets in the views move
- One asset, $\pi = 5\%$, $\tau\sigma^{2} = 0.00128$, view 8% with variance 0.0004: $\mu_{\mathbf{BL}} = 7.29\%$, view weight 0.76
- The same estimator is a mixed regression estimated by generalised least squares (GLS) (Theil & Goldberger (1961); Appendix; Seminar 4, A6)

Black–Litterman on US Sectors



- Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year; implied 2.2%, posterior 2.6% (the average, since Ω equals the prior variance of the view)
- Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged; the view is relative ($P1 = 0$), so the tilt $\propto P^\top$ sums to zero and the portfolio stays fully invested
- Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- Seminar 4, A7 rebuilds the XLK and XLU weights by hand: use them as the check of your result

MFM_ch4_black_litterman

Black–Litterman: What δ_{RA} and τ Do

- $\delta_{\text{RA}} = 2.5$ and $\tau = 0.05$ are the values of He & Litterman (2002)
- **Risk aversion**: premultiply $\boldsymbol{\pi} = \delta_{\text{RA}} \boldsymbol{\Sigma} \mathrm{w}_b$ by $\mathrm{w}_b^\top$

$$\delta_{\text{RA}} = \frac{\mathrm{w}_b^\top \boldsymbol{\pi}}{\mathrm{w}_b^\top \boldsymbol{\Sigma} \mathrm{w}_b} = \frac{\text{benchmark excess return}}{\text{benchmark variance}}$$

  - e.g. 6% excess return at 15.5% volatility: $0.06/0.155^2 = 2.5$
  - $\boldsymbol{\pi}$ is linear in δ_{RA}: doubling it doubles the implied XLK minus XLU spread of 2.2%
- **Prior uncertainty**: $\tau\boldsymbol{\Sigma}$ is the uncertainty about the mean; one reading is $\tau \approx 1/Y$, the variance of a mean estimated over Y years relative to $\boldsymbol{\Sigma}$ (Merton's argument), so $\tau = 0.05$ is about 20 years of data
- With $\boldsymbol{\Omega} = \mathrm{diag}(\mathrm{P}\tau\boldsymbol{\Sigma}\mathrm{P}^\top)$ and one view, τ cancels: the posterior spread is the midpoint $(2.2\% + 3\%)/2 = 2.6\%$, and the weights do not depend on τ
- The confidence in the view is therefore set by $\boldsymbol{\Omega}$ relative to $\tau\boldsymbol{\Sigma}$, not by τ alone

Risk Contributions

- Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^{N} w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^{N} \frac{w_i (\Sigma \mathrm{w})_i}{\sigma_p}$$

- **Risk contribution** of asset i: $RC_i = w_i(\Sigma \mathrm{w})_i/\sigma_p$; percentage share $RC_i/\sigma_p = w_i(\Sigma \mathrm{w})_i/\sigma_p^2$
- Worked example, 50/50 in SPY and IEF (monthly 2002–2026, annualised): $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$
  - $(\Sigma \mathrm{w})_1 = 0.5(0.02166) + 0.5(-0.00098) = 0.01034$, $(\Sigma \mathrm{w})_2 = 0.5(-0.00098) + 0.5(0.00441) = 0.00172$
  - $\sigma_p^2 = 0.5(0.01034) + 0.5(0.00172) = 0.00603$, $\sigma_p = 7.76\%$; shares: SPY 85.8%, IEF 14.2%
- Equal weights are not equal risks: volatile assets dominate; risk budgeting chooses w so that the shares equal preset budgets b_i

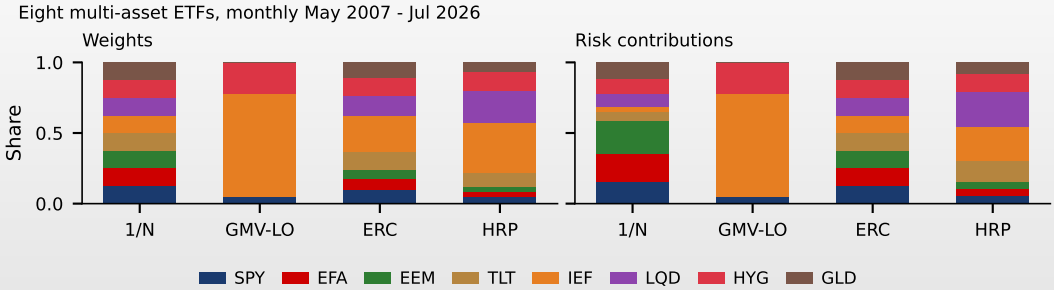
Equal Risk Contribution

- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j; computed from a convex problem, then normalised

$$\min_{\mathrm{y}>0} \ \tfrac{1}{2}\mathrm{y}^\top \boldsymbol{\Sigma}\mathrm{y} - \tfrac{1}{N}\sum_i \ln y_i, \qquad \mathrm{w} = \mathrm{y}/1^\top\mathrm{y}$$

  - y: unnormalised positive weights; the log term keeps every $y_i > 0$ and its first-order condition gives $y_i(\boldsymbol{\Sigma}\mathrm{y})_i = 1/N$ for all i
- For $\boldsymbol{\Sigma}$ positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)
- Volatility ordering: $\sigma_{\mathrm{GMV}} \leq \sigma_{\mathrm{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1}/(\sigma_1^{-1} + \sigma_2^{-1})$, for any $\rho \in (-1, 1)$
  - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY, volatility 6.1%, each asset 50% of the variance
- Needs no expected returns, only $\boldsymbol{\Sigma}$; ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Weights versus Risk Contributions



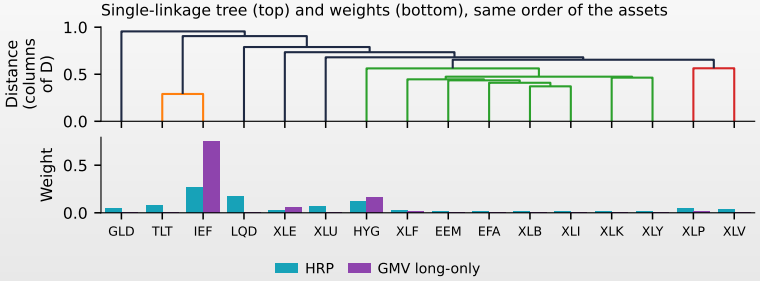
- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$); chart shares in decimals ($0.5 = 50\%$)
- 1/N: equities and high yield receive half of the weight but 69% of the risk; ERC: 12.5% of the risk from every asset, IEF gets the largest weight (25%); long-only GMV: 73% IEF, 22% HYG, 5% SPY, five assets at zero
- Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%: ERC lies between 1/N and GMV, as the theory predicts

MFM_ch4_risk_contrib

Hierarchical Risk Parity: From a Tree to Weights

- HRP (López de Prado (2016)): three steps, no matrix inversion, all weights positive
  - works even when $\boldsymbol{\Sigma}$ is singular
- **Step 1, tree**
  - correlation distance $d_{ij} = \sqrt{(1-\rho_{ij})/2} \in [0,1]$; ρ_{ij}: correlation of assets i and j
  - Euclidean distance between the **columns** of $\mathrm{D} = (d_{ij})$: $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$
  - single linkage on $\tilde{d}_{ij}$ (repeatedly merge the two closest clusters) gives a **dendrogram**
- **Step 2, quasi-diagonalisation**
  - order the assets as the leaves of the tree, so that similar ones are adjacent
- Seminar 4 separates the distances from the tree and so counts four steps; the algorithm is the same
- **Step 3, recursive bisection**: split the **ordered list** in two halves
  - each half gets weight in inverse proportion to its variance $\tilde{\mathbf{w}}^\top \boldsymbol{\Sigma} \tilde{\mathbf{w}}$
  - $\tilde{\mathbf{w}}$: the inverse-variance weights inside the half
  - repeat inside each half until single assets remain
- Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- Textbook: Härdle & Simar (2019), cluster analysis
- A worked four-asset example: Seminar 4, B4

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026; closest pair TLT and IEF ($\tilde{d} = 0.29$), gold joins last (0.96)
- Equity cluster: cyclical sectors, EFA, EEM and high-yield credit; defensive sectors (utilities, staples, health care) and energy join it late
- HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6%, HYG 12.6%; long-only GMV: 75.6% in IEF and only five assets above zero

MFM_ch4_hrp

Backtest Design

- Rolling window of 60 months, monthly rebalancing; the weights of month t use months $t-60, \ldots, t-1$ only
- Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- Three universes
  - Sectors: OOS January 2004 – July 2026, 271 months (first window January 1999 – December 2003)
  - Multi-asset and combined: OOS May 2012 – July 2026, 171 months
- Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth
- Every Sharpe ratio difference is reported with a test (next slides), and every test run enters the multiple-testing correction

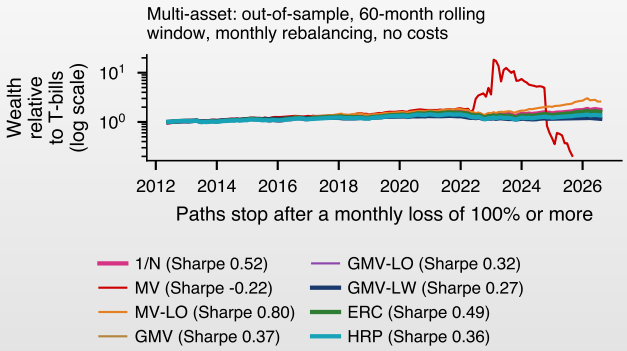
Turnover and Costs: One Worked Month (1/2)

- Between rebalancings weights drift with returns: $w_{i,t}^{+} = w_{i,t}(1 + R_{i,t}) / \sum_j w_{j,t}(1 + R_{j,t})$
  - $R_{i,t}$: **total** return of asset i in month t; the denominator is $1 + R_{p,t}$, one plus the portfolio's total return
- **Turnover**: traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^{+}|$, summed over all N assets
- Worked month, sectors, 1/N: in January 2004 XLB returned -4.4% and XLK $+4.3\%$, so the weights of 11.11% drifted to 10.50% and 11.45%
  - 1/N total return in January 2004: 1.18% (excess 1.11% plus the Treasury bill 0.07%; Seminar 4, B1)
  - XLB: $\frac{1}{9}(1 - 0.044)/1.0118 = 10.50\%$; XLK: $\frac{1}{9}(1 + 0.043)/1.0118 = 11.45\%$

Turnover and Costs: One Worked Month (2/2)

- Rebalancing back to 1/N in February 2004 trades $TO = 0.019$: 1/N also trades, selling winners and buying losers
  - XLB and XLK contribute $|0.1111 - 0.1050| + |0.1111 - 0.1145| = 0.0061 + 0.0034 = 0.0095$; the other seven sectors about 0.009 together (apply the same drift formula to the other seven sector returns in Seminar 4, B1)
  - MV-LO, re-estimated on the new window, trades 0.748 in the same month
- Net return with cost c per unit traded: $R^{\text{net}}_{p,t} = R_{p,t} - c\, TO_t$; 1 basis point (bp) $= 0.01\%$, so 20 bp means $c = 0.002$
  - At 20 bp the February trade costs MV-LO $0.002 \times 0.748 = 0.15\%$ of wealth, 1/N $0.002 \times 0.019 = 0.004\%$

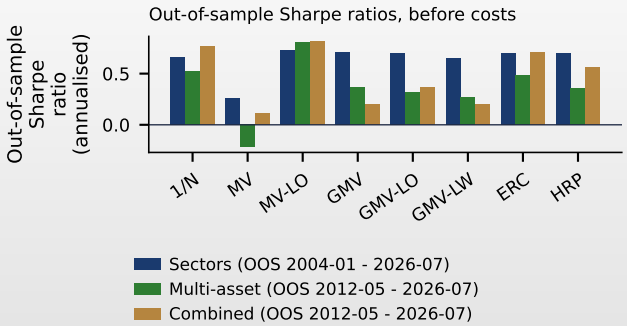
Out-of-Sample Growth: Multi-Asset



- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs
- MV collapses: volatility 518% a year, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- GMV rules: lowest volatility (3.8%–5.4%), low Sharpe (0.27–0.37); they load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT −0.5%, IEF −0.2% a year)
- Is MV-LO really better, or lucky? A test is needed

MFM_ch4_oos_backtest

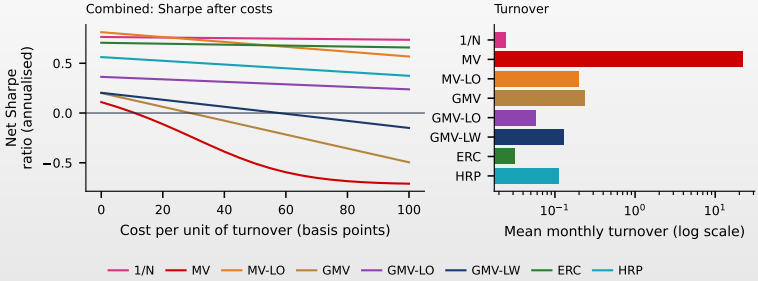
Sharpe Ratios Across Universes



- 60-month rolling window, monthly rebalancing, annualised, before costs; complete tables for all three universes in the ▸ Appendix
- Sectors: every rule except MV lands within 0.07 of 1/N (0.66); MV 0.26 with drawdown −100%
- This 0.66 is OOS 2004–2026; the 0.54 (frontier, simulations) is 1/N on 1999–2026: different periods, so matching the true tangency 0.66 does not show that 1/N is efficient
- Combined: 1/N 0.76, MV-LO 0.81, ERC 0.71; the GMV rules 0.20–0.36
- Turnover per month, sectors: 1/N 0.02, ERC 0.03, GMV-LW 0.14, MV-LO 0.23, MV 3.86

MFM_ch4_oos_backtest

Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover; turnover from 0.02 (1/N) to 22 (MV) per month
- Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW; MV-LO overtakes 1/N only for costs below about 22 bp (Seminar 4, B2 derives this break-even; 22 bp is its check value)
- 1/N and ERC barely move: low turnover is a hidden advantage; report net performance for realistic costs, not only gross

MFM_ch4_costs

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \to N\big(0,\ 1 + SR^2/2\big)$, with SR monthly, T the number of months: the standard error is $\sqrt{(1 + SR^2/2)/T}$
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
  - Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{A}$ ($A = 12$ months per year; Lo writes q, and this A is not the $1^\top \boldsymbol{\Sigma}^{-1} 1$ of the GMV) is exact only without autocorrelation; in general $SR(A) = \eta(A)\, SR$

$$\eta(A) = \frac{A}{\sqrt{A + 2\sum_{k=1}^{A-1}(A-k)\rho_k}}$$

  - ρ_k: autocorrelation of monthly returns at lag k; positive ρ_k lower η, negative ρ_k raise it; $\rho_k = 0$ gives $\eta = \sqrt{A}$
- Here $\rho_1 = -0.03$, $\sum_{k=1}^{11} \rho_k = -0.11$; the formula needs the weighted sum $\sum_{k=1}^{11}(12-k)\rho_k = -0.50$
  - $\eta = 12/\sqrt{12 - 2(0.50)} = 12/\sqrt{11.0} = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe $0.191 \times 3.62 = 0.69$

Testing a Difference in Sharpe Ratios

- H_0: $SR_1 = SR_2$ for two strategies evaluated on the same months; Jobson & Korkie (1981): a test under i.i.d. Normal returns
- Ledoit & Wolf (2008): a robust version, by the delta method

$$\hat{\Delta} = \widehat{SR}_1 - \widehat{SR}_2 = f(\hat{\mathrm{v}}), \qquad f(\mathrm{v}) = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$$

  - r_{1t}, r_{2t}: excess returns of the two strategies; μ_1, μ_2: their means; $\mathrm{v} = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$
  - standard error $\sqrt{\nabla f^\top \hat{\boldsymbol{\Psi}} \nabla f / T}$; ∇f: the gradient of f
  - $\hat{\boldsymbol{\Psi}}$: a HAC (heteroskedasticity and autocorrelation consistent) covariance of the four moments
- Strategies on the same assets are highly correlated: small differences can be significant

The HAC Test Step by Step

- Gradient of f with respect to $\mathrm{v} = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, writing $\sigma_i^2 = \mathbb{E}[r_i^2] - \mu_i^2$

$$\nabla f = \Big(\frac{\mathbb{E}[r_1^2]}{\sigma_1^3},\ -\frac{\mathbb{E}[r_2^2]}{\sigma_2^3},\ -\frac{\mu_1}{2\sigma_1^3},\ \frac{\mu_2}{2\sigma_2^3} \Big)$$

  - these are derivatives in the **uncentred** second moments; a gradient in (μ_i, σ_i^2) does not fit this $\hat{\Psi}$

- $\hat{\Psi}$: long-run covariance of $\mathrm{y}_t = (r_{1t} - \hat{\mu}_1,\ r_{2t} - \hat{\mu}_2,\ r_{1t}^2 - \widehat{\mathbb{E}r_1^2},\ r_{2t}^2 - \widehat{\mathbb{E}r_2^2})$

- HAC p-values in this chapter's tables: Bartlett (Newey–West) kernel with $\lfloor 4(T/100)^{2/9} \rfloor = 4$ lags for every T from 107 to 271; $z = \hat{\Delta}/\mathrm{SE}$, $p = 2[1 - \Phi(|z|)]$

- The bootstrap studentizes with a different SE (prewhitened quadratic-spectral kernel, Appendix), so the two statistics differ slightly
  - sectors, MV-LO minus 1/N: $\hat{\Delta} = 0.063$; Bartlett SE 0.120, $z = 0.53$, HAC $p = 0.60$; QS SE 0.127, studentized statistic 0.50, the observed value on the bootstrap chart of Seminar 4, B1

The Block Bootstrap

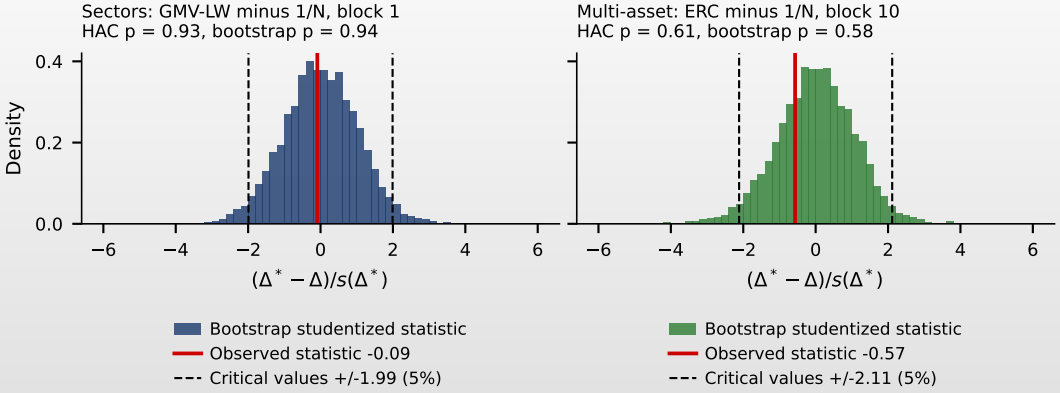


Bradley Efron, inventor of the bootstrap (1979)

Photo: Ryan K. Morris (2007); public domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of **pairs** (r_{1t}, r_{2t}), wrapping around the end of the sample, so the correlation of the two strategies is kept
- Centre and studentize (divide by an estimated standard error): bootstrap $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$; asymptotically pivotal (its limit distribution has no unknown parameters), more accurate than the Normal approximation of the HAC test
- $p = \left[\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1\right]/(M+1)$, $M = 4999$ draws; $\tilde{d}^{*m}$: the studentized statistic in draw m, $s(\cdot)$: standard error (SE in the seminar), $\#\{\cdot\}$: number of draws
- Block length chosen by the calibration of Ledoit & Wolf (2008), Algorithm 3.1 (▸ Appendix): 1 for six of the seven sector rules; between 1 and 10, most often 10, for multi-asset and combined rules (every block in the 21-test table of the Appendix)

Is Anything Better than 1/N? (1/2)



MFM_ch4_sharpe_test

Is Anything Better than 1/N? (2/2)

- Bootstrap distributions of the studentized difference against 1/N; red: observed statistic; dashed: bootstrap critical values
- Sectors, GMV-LW minus 1/N: -0.02, 95% interval $[-0.36, 0.33]$, HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1); multi-asset, ERC minus 1/N: -0.03, $[-0.15, 0.08]$, HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10): so correlated with 1/N that the interval is narrow, yet it covers zero
- Sectors: no rule differs from 1/N at 5% (closest ERC, $+0.04$, HAC $p = 0.09$); multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

MFM_ch4_sharpe_test

Many Rules, Many Tests

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.05
Combined	GMV	-0.56	0.03	0.05
Combined	GMV-LO	-0.40	0.05	0.08
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	$+0.05$	0.82	0.82

- No rule beats 1/N significantly in any universe; unadjusted, GMV rules lose to 1/N on the combined universe: low risk, too little return
- The table shows 6 of the 21 two-sided tests, H_0: $\Delta_k = 0$ (7 rules $\times$ 3 universes; all 21 in the ▸ Appendix). Holm over all 21: smallest adjusted p is 0.29 (HAC, 21×0.014) and 0.68 (bootstrap, 21×0.032): nothing survives (next slide; Chapter 3)
- The separate superiority question (one-sided): H_0: $\Delta_k = SR_k - SR_{1/N} \leq 0$ for all k; the same 21 pairs
- A different question, the best of 24 configurations: the deflated Sharpe ratio (DSR) of Bailey & López de Prado (2014) gives $0.84 < 0.95$ for MV-LO on 16 ETFs, but the verdict hinges on the effective number of trials (▸ Appendix)

Holm and Benjamini–Hochberg on This Chapter's Tests

- **FWER** (family-wise error rate): the probability of at least one false rejection in the family; **FDR** (false discovery rate): the expected share of false rejections among the rejections
- Sort the m p-values, $p_{(1)} \leq \cdots \leq p_{(m)}$
  - **Holm** (controls the FWER under any dependence): $\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (m - j + 1)\, p_{(j)}\}$
  - **Benjamini–Hochberg** (Benjamini & Hochberg (1995); controls the FDR): $\tilde{p}_{(i)} = \min_{j \geq i} \min\{1, \frac{m}{j}\, p_{(j)}\}$
  - reject where $\tilde{p} < 0.05$ (Holm) or $\tilde{p} <$ the chosen FDR (BH)
- The four smallest two-sided HAC p-values, $m = 21$

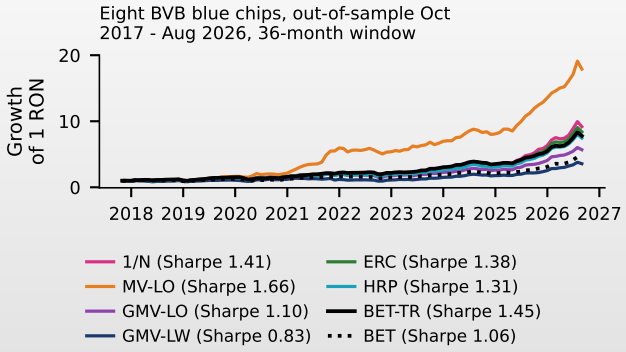
Test	$p_{(j)}$	Holm	$\frac{21}{j}\, p_{(j)}$
Multi-asset, MV	0.0138	$21 \times 0.0138 = 0.289$	0.289
Combined, GMV-LW	0.0238	$20 \times 0.0238 = 0.475$	0.249
Combined, GMV	0.0304	$19 \times 0.0304 = 0.577$	0.213
Combined, GMV-LO	0.0477	$18 \times 0.0477 = 0.858$	0.250

- BH takes the minimum from the bottom up: the first three all become 0.213; no rejection at an FDR of 10%

Setup: a Romanian Investor

- Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR



- Out of sample, October 2017 – August 2026 (107 months), before costs, $R_f = 0$
- BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.06; the gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- Rule minus BET-TR: MV-LO 1.66, $+0.21$ (HAC $p = 0.44$, bootstrap 0.53); GMV-LW 0.83, -0.63 (HAC $p = 0.05$, bootstrap 0.10)
- Average correlation of the eight stocks 0.44: little room for diversification inside the BVB; the latest GMV-LO puts 70% in TLV
- Open question: does diversification **beyond** the BVB pay? (Seminar 4, C2)

MFM_ch4_bvb

Which Frontier Can a Large Investor Reach? (1/2)

- Deeper study: the idea used later is only that strategies are ranked by the frontier **net of trading costs**
- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*
  - how should portfolios be built from machine-learning (ML) return forecasts when every trade moves prices?
- **Implementable efficient frontier** (eq. 12): for each volatility σ, the highest expected return **net of trading costs**

$$k(\sigma) = \max_{\pi} \; E[r_{t+1}^{\text{net}}] \quad \text{s.t.} \quad E[\pi_t^\top \boldsymbol{\Sigma} \pi_t] = \sigma^2$$

  - π_t: vector of risky-asset weights at t (the paper's notation; not the implied returns $\boldsymbol{\pi}$ of Black–Litterman); r_{t+1}^{net}: portfolio return after trading costs
  - $k(\sigma)$: the best net expected return at volatility σ
- Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier
  - a larger investor faces a lower frontier

Which Frontier Can a Large Investor Reach? (2/2)

- Cost as a fraction of wealth W_t (eqs. 4, 7), quadratic in the trade

$$c_t = \tfrac{W_t}{2}(\pi_t - g_t\pi_{t-\mathbf{1}})^\top \mathbf{\Lambda}(\pi_t - g_t\pi_{t-\mathbf{1}}), \qquad \mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$$

  - g_t: drifts last month's weights; V_i: the dollar volume of asset i
- Optimal rule: trade only part of the way towards an **aim portfolio** A_t (Gârleanu & Pedersen (2013))

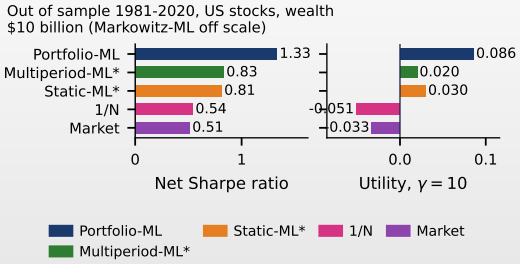
$$\pi_t = m\, g_t\pi_{t-\mathbf{1}} + (\mathbf{I} - m)A_t$$

  - A_t: the portfolio one would hold if trading were free, a weighted average of today's and expected future Markowitz portfolios
  - m: an $N \times N$ matrix of trade rates (eq. 14), so $\mathbf{I} - m$ is a matrix too; in the scalar special case of Gârleanu & Pedersen (2013), m is the share of the old position kept each period
  - Portfolio-ML learns A_t from the signals by a ridge regression (least squares with a penalty on the size of the coefficients) whose objective is utility net of costs

Paper and Course Application: What Is Compared?

	Paper	**Course replication**
Assets	NYSE (New York Stock Exchange) stocks above median size (184–805)	14 US stocks + 9 sector ETFs
Signals	115 characteristics (Jensen, Kelly & Pedersen (2023))	5 price and volume signals
Out of sample	1981–2020, monthly	February 2014 – August 2026 (151 months)
Investor	$\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market	same; wealth scaled by the cumulative SPY return before and after 2020
Utility	annual net return $-\frac{\gamma}{2}\times$ annual variance	same: $12\times$ monthly mean $-\frac{\gamma}{2}\times 12\times$ monthly variance

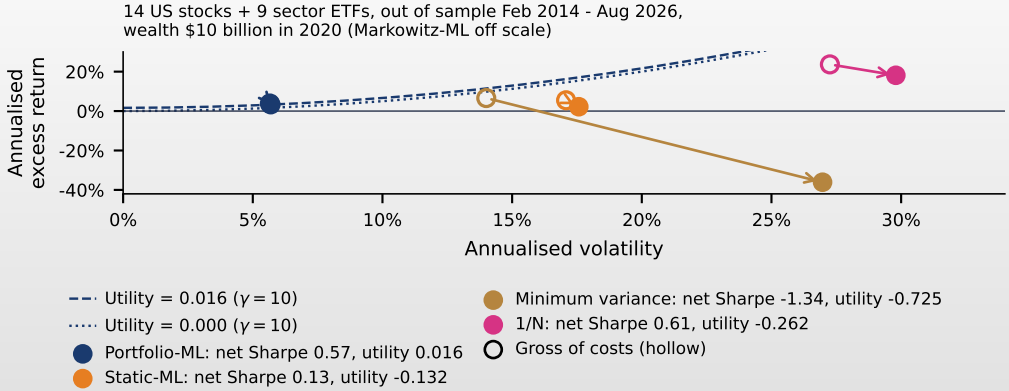
Table 2 at a Glance



- Jensen, Kelly, Malamud & Pedersen (2026), Table 2: net of costs, 1981–2020, $\gamma = 10$; utility (eq. 11) = annual net return $-\frac{\gamma}{2}\times$ annual variance
- Markowitz-ML (mean–variance on ML return forecasts, no costs): gross Sharpe 2.06, turnover 79.36, leverage 74.61: net return extremely negative
- Portfolio-ML: net Sharpe 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year)
- Multiperiod-ML* (forecasts for 12 horizons): net Sharpe 0.83, utility 0.020
- Static-ML* (one-period costs; * = extra tuning layer) 0.81, 0.030; 1/N 0.54, −0.051; market 0.51, −0.033: only cost-aware rules have positive utility

MFM_ch4_implementable_frontier

Gross and Net of Costs on the Course Data (1/2)



MFM_ch4_implementable_frontier

Gross and Net of Costs on the Course Data (2/2)

- Hollow marker: before costs; filled: after the quadratic cost c_t with $\mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$ (eq. 35 of the paper); dashed curve: pairs with the utility of Portfolio-ML (0.016), dotted: utility 0
- Portfolio-ML barely moves: 3.8% gross, 3.3% net, 5.7% volatility; net Sharpe 0.57
  - utility, annual units: $3.26\% - \frac{10}{2} \times (5.69\%)^2 = 0.0326 - 0.0162 = 0.016$ (net mean and net volatility)
- 1/N keeps 18.1% net (Sharpe 0.61), but at 29.8% volatility its utility is -0.26
- Minimum variance: $+6.6\%$ gross, -36.1% net, rebalancing a leverage of 2.6 costs 43% a year
- Limits: 23 assets, 5 signals, no factor model for $\mathbf{\Sigma}$; the frontier that matters is the one after costs (Seminar 4, C3)

MFM_ch4_implementable_frontier

Open Questions

- Questions
  - does any **estimated** allocation rule beat 1/N out of sample, net of costs?
  - does it still beat 1/N once every rule, window and universe tried is counted?
- Live evidence points both ways
  - Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: Ledoit & Wolf (2017)
  - Combining tangency, GMV and 1/N under estimation risk: Kan, Wang & Zhou (2022)
  - Backtests need haircuts for the number of specifications tried: Harvey & Liu (2015)
- Formal statement: the family H_0 of Many Rules, Many Tests against H_1: $\Delta_k > 0$ for some k, with the error rate fixed for the whole family
- One pass of the loop with an AI assistant: literature with DOIs, competing hypotheses, code on the course's monthly data, a pre-registered family of tests, a referee-style critique (prompts in the Appendix)
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

What the Researcher Must Check

- **Look-ahead**: weights at t built with month-t returns
  - test: assert that the last estimation date is before t
- **Survivorship and selection**: an ETF list chosen today
  - test: use the universe available at each date, including later delisted ETFs, or disclose the limitation
- **Multiple testing**: 7 rules $\times$ 3 universes already give 21 tests
  - test: Holm-adjusted p-values, which control the FWER (family-wise error rate, the chance of at least one false rejection)
  - or Benjamini–Hochberg (Benjamini & Hochberg (1995)), which controls the FDR (false discovery rate, the expected share of false rejections)
- **Power**: $\hat{\Delta}$ has a standard error near 0.18 on 171 months
  - test: compute the detectable effect before running
- **Return type**: portfolios aggregate **simple** returns, Sharpe ratios use **excess** returns
  - test: recompute one month step by step
- **Invented references**: an unresolvable DOI or a title mismatch
  - the claim is dropped and the error goes into `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest (1/2)

- A typical assistant proposal: "MV-LO beats 1/N on the multi-asset universe"; superiority is one-sided: H_0: $\Delta \leq 0$, $p = 1 - \Phi(\hat{\Delta}/\mathrm{SE})$, Φ the standard Normal distribution function, SE the standard error
- Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$ (two-sided HAC 0.12, bootstrap 0.16)
  - recompute from four decimals, not from the rounded values: $z = 0.2863/0.1836 = 1.56$, $p = 1 - \Phi(1.56) = 0.06$; the rounded $0.29/0.18 = 1.61$ would give 0.054
- Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
  - months needed: $T_{\mathrm{obs}}\left[(z_{1-\alpha} + z_{0.8})\,\mathrm{SE}/\Delta\right]^2 = 171\left[(1.645 + 0.842) \times 0.1836/0.2863\right]^2 \approx 435$
  - at $\alpha = 0.05/21$, $z_{1-\alpha} = 2.82$, which gives ≈ 944

MFM_ch4_ai_discovery

Mini-Case: Data Snooping in This Chapter's Backtest (2/2)

- Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes; one-sided p-values of all 21 in the Appendix)
  - Only 7 of 21 differences are positive; the 4 two-sided HAC p-values below 0.05 are all **losses**, which never reject a superiority null
  - One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- Why 0.96 here but 0.29 on Many Rules, Many Tests: a loss has a large one-sided p (about $1 - p_{\text{two-sided}}/2$, e.g. 0.99 for multi-asset MV), so the significant losses drop out; the smallest one-sided p belongs to a gain, 0.046, and Holm multiplies it by 21
- Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains; failure to reject is not proof of equality

MFM_ch4_ai_discovery

Your Turn: A Research Project Idea

- **Questions**
  - does nonlinear shrinkage (Ledoit & Wolf (2017)) beat 1/N on the 16 ETFs of this chapter, after 10–50 bp costs?
  - does a three-fund combination (Kan, Wang & Zhou (2022)) beat 1/N on the same ETFs, after the same costs?
- **Replicate first**: the combined-universe results (1/N Sharpe 0.76, MV-LO 0.81) and the adjusted *p*-values of the mini-case
- **Pre-register**, before any result
  - hypotheses, family of tests, error rate and power
  - a placebo: random long-only weights with the same turnover
- **Report**: code on the course's monthly data, a table of all specifications, a referee-style self-critique
- **Log**: prompts in `AI_USE.md`; every error caught and the test that caught it in `AI_ERRORS.md`

Conclusions: Which Rule, When?

Rule	Objective	Inputs	Main failure mode	Evidence in this chapter
MV	maximum Sharpe	$\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}^{-1}$	error maximisation, explosive weights	worst in every universe; insolvent path
MV-LO	maximum Sharpe, $w_i \geq 0$	$\hat{\boldsymbol{\mu}}$, $\hat{\boldsymbol{\Sigma}}$	corner solutions, turnover	best point estimates, never significant
GMV, GMV-LW	minimum variance	$\hat{\boldsymbol{\Sigma}}$ (shrunk)	loads on low-return assets	lowest risk, loses to 1/N on 16 ETFs (unadjusted)
ERC, HRP	balanced risk	$\hat{\boldsymbol{\Sigma}}$	a tilt of 1/N; HRP trades more	never significantly better than 1/N
Black–Litterman	benchmark plus views	w_b, views, $\boldsymbol{\Omega}$	confidence set too high	moves only the assets in the views
1/N	none	none	ignores risk differences	hard to beat after costs

- Evaluate every rule out of sample, net of costs, with a test of the Sharpe difference and a count of all tests; these verdicts are specific to these samples

Quick Checks with Toy Numbers (1/2)

- **Three rules, two assets**: $\sigma_1 = 30\%$, $\sigma_2 = 15\%$, $\rho = 0.2$
  - compute the weights and volatilities of GMV, ERC and 1/N; check $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
  - what share of the 1/N variance comes from asset 1? (Two Assets; Risk Contributions; Equal Risk Contribution)
- **Optimism of the plug-in Sharpe**: $N = 4$ assets, $T = 120$ months, true $\theta = 0.50$ a year
  - what in-sample maximum Sharpe (root mean square, annual) should you expect? (The Plug-in Sharpe Ratio Is Biased)
- **Noise band**: $N = 10$, $T = 40$, correlation matrix
  - compute the Marchenko–Pastur band; is a sample eigenvalue of 2.0 evidence of structure? (Why the Sample Covariance Matrix Fails)

Quick Checks with Toy Numbers (2/2)

- **One view, one asset**: $\pi = 4\%$, $\tau\sigma^2 = 0.0009$, view 10%
  - posterior mean and view weight for $\Omega = 0.0009$ and for $\Omega = 0.0003$ (Views and the Posterior)
- **One Sharpe ratio**: monthly $\widehat{SR} = 0.15$, $T = 120$, i.i.d. Normal
  - annual Sharpe and its 95% interval: is it significantly positive? (Inference on One Sharpe Ratio)
- **Four tests**: $p = 0.004,\ 0.020,\ 0.030,\ 0.300$
  - Holm and Benjamini–Hochberg adjusted p-values; how many rejections at 5% under each? (Holm and Benjamini–Hochberg)
- Answers: Appendix

Check Yourself, and Next

Retrieval questions

- Why does the in-sample tangency Sharpe overstate the achievable one? (plug-in Sharpe; Seminar 4, A8)
- By how much does it overstate it for $N = 9$, $T = 60$? (same slide)
- Why does a long-only constraint act like shrinkage? (Jagannathan–Ma and its two-asset example)
- Why can a small Sharpe difference between ERC and 1/N be nearly significant while a larger one between GMV and 1/N is not? (HAC test: compare the standard errors 0.023 and 0.159 in the Appendix table)

Links to other chapters

- **Chapter 3**: the tangency portfolio and the CAPM; factor covariance models
- **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- **Chapter 13**: backtest overfitting
- **Chapter 13**: machine learning for returns and portfolios

The Frontier in Closed Form, Step by Step

- Two constraints, two multipliers: $\mathcal{L} = \frac{1}{2}\mathrm{w}^\top \boldsymbol{\Sigma}\mathrm{w} - \lambda_1(\mathrm{w}^\top \boldsymbol{\mu} - m) - \lambda_2(\mathrm{w}^\top 1 - 1)$
- First-order condition: $\boldsymbol{\Sigma}\mathrm{w} = \lambda_1\boldsymbol{\mu} + \lambda_2 1$, so $\mathrm{w} = \lambda_1\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \lambda_2\boldsymbol{\Sigma}^{-1}1$
- The constraints, with $A = 1^\top\boldsymbol{\Sigma}^{-1}1$, $B = 1^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $D = AC - B^2$

$$\boldsymbol{\mu}^\top \mathrm{w} = \lambda_1 C + \lambda_2 B = m, \qquad 1^\top \mathrm{w} = \lambda_1 B + \lambda_2 A = 1$$

- Solving the 2×2 system: $\lambda_1 = (Am - B)/D$, $\lambda_2 = (C - Bm)/D$
- Variance: $\mathrm{w}^\top\boldsymbol{\Sigma}\mathrm{w} = \mathrm{w}^\top(\lambda_1\boldsymbol{\mu} + \lambda_2 1) = \lambda_1 m + \lambda_2 = \dfrac{Am^2 - 2Bm + C}{D}$
- Check: at $m = B/A$ the variance is $(C - B^2/A)/D = 1/A$, the GMV
- Shapes: σ^2 is quadratic in m (a parabola in the (σ^2, m) plane); $\sigma^2 - A(m - B/A)^2/D = 1/A$ is a hyperbola in the (σ, m) plane

Back

Wishart Moments and the Distribution of Weights

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood estimators: $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$ (Wishart: the matrix analogue of χ^2, $T-1$ degrees of freedom), independent
- Inverse Wishart moment, finite for $T > N+2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$, an **upward** bias that grows with N/T
- Conditioning on $\hat{\boldsymbol{\mu}}$ and using $\mathbb{E}[\mathrm{x}^\top \mathrm{K}\mathrm{x}] = \mathrm{tr}(\mathrm{K}\,\mathrm{Var}(\mathrm{x})) + \mathbb{E}[\mathrm{x}]^\top \mathrm{K}\,\mathbb{E}[\mathrm{x}]$ (any random vector x, fixed matrix K, tr: trace) gives $\mathbb{E}[\hat{\theta}^2]$ of the lecture (Seminar 4, A8)
- GMV weights with Wishart $\hat{\boldsymbol{\Sigma}}$ and T large enough relative to N: $\hat{\mathrm{w}}_{\mathrm{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $1^\top \hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\mu}}$, which can be near zero: very heavy tails
- Estimation-risk-optimal mixtures of tangency, GMV and 1/N: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

Back

Tangency Weights by Regression, Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathrm{b}^\top \mathrm{r}_t + u_t, \qquad \hat{\mathrm{b}} = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top 1_T \propto \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$$

- r_t: vector of excess returns in month t; X: $T \times N$ matrix with rows $\mathrm{r}_t^\top$; 1_T: vector of T ones; u_t: regression error
- Sherman–Morrison on $\mathrm{X}^\top \mathrm{X}/T = \hat{\boldsymbol{\Sigma}} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar 4, A9); $\hat{\mathrm{b}}/1^\top \hat{\mathrm{b}}$ = sample tangency weights if $1^\top \hat{\mathrm{b}} > 0$
- OLS t-test of $b_i = 0$: asset i has zero weight in the tangency portfolio; F-test of $\mathrm{Rb} = 0$ with $\mathrm{R} = [\mathrm{I}_{N-1}, -1_{N-1}]$: "the tangency portfolio is 1/N"
- Exact finite-sample t and F distributions under i.i.d. Normal returns; HAC standard errors otherwise
- Nine sectors, January 1999 – July 2026 ($T = 331$): normalised $\hat{\mathrm{b}}$ reproduces the tangency weights; largest $|t| = 1.12$ (financials), HAC 1.19
- H_0: tangency = 1/N, eight restrictions: $F(8, 322) = 0.48$, $p = 0.87$; HAC Wald $\chi^2_8 = 4.50$, $p = 0.81$; not rejecting is not proof of equal weights (replication target of Seminar 4, B6, full sample)

Back

The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true $\boldsymbol{\Sigma}$

$$\delta^* = \max\left\{0, \min\left\{1, \frac{\hat{\kappa}}{T}\right\}\right\}, \qquad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their asymptotic covariances with the target entries
- $\hat{\gamma} = \|\mathsf{F} - \mathsf{S}\|^2$: misspecification of the target (not the risk aversion γ)
- Complete formulas for the constant-correlation target: Ledoit & Wolf (2004b), Appendix; code: the shrinkage Quantlet

Back

Shrinking the Means and Other Covariance Estimators

- Bayes–Stein (Jorion (1986)): $\hat{\boldsymbol{\mu}}_{\mathrm{BS}} = (1 - \hat{\phi})\,\hat{\boldsymbol{\mu}} + \hat{\phi}\,\hat{\mu}_{\mathrm{GMV}}\mathbf{1}$
  - Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- Tu & Zhou (2011): the convex combination of a sophisticated rule and 1/N that minimises the expected out-of-sample loss
- DeMiguel, Garlappi, Nogales & Uppal (2009): norm constraints $\|\mathrm{w}\|_1 \leq \delta$ or $\|\mathrm{w}\|_2 \leq \delta$ nest the long-only GMV, shrinkage GMV and 1/N; the bridge to the lasso portfolios of Chapter 13
- Factor structure plus a sparse residual covariance: POET (principal orthogonal complement thresholding, Fan, Liao & Mincheva (2013)), the link to Chapter 3
- High-dimensional GMV with shrinkage towards a target portfolio: Bodnar, Parolya & Schmid (2018)

Back

Black–Litterman as Mixed (GLS) Estimation

- Theil & Goldberger (1961): stack the prior and the views as one regression on $\boldsymbol{\mu}$

$$\begin{pmatrix} \boldsymbol{\pi} \\ \mathsf{q} \end{pmatrix} = \begin{pmatrix} \mathsf{I} \\ \mathsf{P} \end{pmatrix} \boldsymbol{\mu} + \begin{pmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{pmatrix}, \qquad \operatorname{Var} \begin{pmatrix} \mathsf{e}_1 \\ \mathsf{e}_2 \end{pmatrix} = \begin{pmatrix} \tau\boldsymbol{\Sigma} & 0 \\ 0 & \boldsymbol{\Omega} \end{pmatrix}$$

- GLS gives exactly $\boldsymbol{\mu}_{\mathsf{BL}}$, with covariance $\mathsf{M} = \left[(\tau\boldsymbol{\Sigma})^{-1} + \mathsf{P}^\top \boldsymbol{\Omega}^{-1} \mathsf{P}\right]^{-1}$
- Equivalent form: $\boldsymbol{\mu}_{\mathsf{BL}} = \boldsymbol{\pi} + \tau\boldsymbol{\Sigma}\mathsf{P}^\top (\mathsf{P}\tau\boldsymbol{\Sigma}\mathsf{P}^\top + \boldsymbol{\Omega})^{-1}(\mathsf{q} - \mathsf{P}\boldsymbol{\pi})$
- $\boldsymbol{\Omega} \to 0$: restricted least squares, $\mathsf{P}\boldsymbol{\mu}_{\mathsf{BL}} = \mathsf{q}$
- Predictive covariance $\boldsymbol{\Sigma} + \mathsf{M}$ in the weights would change all weights; without views it scales them by $1/(1+\tau)$

Back

Other Risk-Based Rules

- **Inverse volatility**: $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $\mathrm{w}^\top \boldsymbol{\sigma} / \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}$, the ratio of the average asset volatility to the portfolio volatility; $\boldsymbol{\sigma}$: vector of asset volatilities
- **Minimum variance**: GMV, the extreme risk-based rule
- All ignore expected returns: robust to estimation error, but optimal only under special assumptions

Back

Results, Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−49%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−42%
GMV	8.5%	11.9%	0.71	0.23	−33%
GMV-LO	8.1%	11.5%	0.70	0.09	−34%
GMV-LW	7.6%	11.8%	0.65	0.14	−31%
ERC	9.3%	13.2%	0.70	0.03	−47%
HRP	8.9%	12.8%	0.70	0.08	−45%

- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth

Back

Results, Multi-Asset Universe (8 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown	Sharpe, 50 bp
1/N	4.3%	8.4%	0.52	0.02	−21%	0.50
MV	−113.0%	517.5%	−0.22	44.46	−100%	−0.54
MV-LO	7.1%	8.9%	0.80	0.12	−24%	0.72
GMV	1.4%	3.8%	0.37	0.14	−11%	0.16
GMV-LO	1.7%	5.4%	0.32	0.05	−17%	0.27
GMV-LW	1.1%	4.2%	0.27	0.09	−13%	0.15
ERC	3.6%	7.3%	0.49	0.02	−21%	0.47
HRP	2.4%	6.6%	0.36	0.04	−20%	0.32

- May 2012 – July 2026 (171 months); MV's mean and volatility are arithmetic averages of monthly returns that include a loss above 100%, so only its drawdown has an economic meaning

Back

Results, Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	−0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	−0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	4.0%	7.0%	0.56	0.11	0.47

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover

Back

All 21 Tests against 1/N (1/3), Sectors

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.40	0.331	0.22	1.00	0.889	0.30	6
MV-LO	+0.06	0.120	0.60	1.00	0.299	0.63	1
GMV	+0.05	0.159	0.75	1.00	0.377	0.76	1
GMV-LO	+0.04	0.112	0.73	1.00	0.363	0.73	1
GMV-LW	−0.02	0.166	0.93	1.00	0.537	0.94	1
ERC	+0.04	0.023	0.09	1.00	0.046	0.13	1
HRP	+0.04	0.040	0.32	1.00	0.162	0.36	1

- January 2004 – July 2026, $T = 271$; $\hat{\Delta}$: annualised Sharpe of the rule minus that of 1/N, before costs
- HAC: Bartlett, 4 lags, two-sided; Holm over all 21 two-sided HAC tests; one-sided: $1 - \Phi(\hat{\Delta}/\mathrm{SE})$; bootstrap: studentized circular block bootstrap, two-sided, block from Algorithm 3.1

Back

All 21 Tests against 1/N (2/3), Multi-Asset

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	-0.74	0.300	0.01	0.29	0.993	0.06	10
MV-LO	$+0.29$	0.184	0.12	1.00	0.059	0.16	10
GMV	-0.15	0.226	0.51	1.00	0.745	0.54	1
GMV-LO	-0.20	0.137	0.14	1.00	0.930	0.15	8
GMV-LW	-0.25	0.191	0.19	1.00	0.903	0.20	4
ERC	-0.03	0.060	0.61	1.00	0.694	0.58	10
HRP	-0.16	0.089	0.07	1.00	0.964	0.05	6

- May 2012 – July 2026, $T = 171$; same columns as for the sectors
- Check one row: MV-LO, $z = 0.2863/0.1836 = 1.56$; two-sided $p = 2[1 - \Phi(1.56)] = 0.12$, one-sided 0.059

Back

All 21 Tests against 1/N (3/3), Combined

Rule	$\hat{\Delta}$	HAC SE	p HAC	Holm	p one-sided	p bootstrap	Block
MV	−0.66	0.355	0.06	1.00	0.968	0.22	10
MV-LO	+0.05	0.210	0.82	1.00	0.409	0.82	10
GMV	−0.56	0.261	0.03	0.58	0.985	0.05	1
GMV-LO	−0.40	0.203	0.05	0.86	0.976	0.08	6
GMV-LW	−0.56	0.249	0.02	0.48	0.988	0.03	2
ERC	−0.06	0.114	0.61	1.00	0.695	0.62	10
HRP	−0.20	0.152	0.18	1.00	0.908	0.20	8

- May 2012 – July 2026, $T = 171$; 7 of the 21 $\hat{\Delta}$ are positive, and every two-sided HAC $p < 0.05$ is a loss
- One-sided family: Holm $21 \times 0.046 = 0.96$; Benjamini–Hochberg $\min\{21 \times 0.046,\ \frac{21}{2} \times 0.059\} = 0.62$ for the two smallest

Back

Bootstrap Details and the Block Length

- Studentization: $s(\hat{\Delta}^*)$ from the block sums of each resample; $s(\hat{\Delta})$ on the data from a prewhitened quadratic-spectral kernel (Andrews & Monahan (1992)) with automatic bandwidth (Andrews (1991))
  - the HAC p-values in the tables use the simpler Bartlett kernel with 4 lags (HAC test); the two SEs differ mostly in the second decimal
- Ledoit & Wolf (2008), Algorithm 3.1: choose the block b whose bootstrap interval has actual coverage closest to 95%
  - Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap (Politis & Romano (1994)), mean block 5
  - Generate $K = 5000$ pseudo-samples of length T; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
  - $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg\min_b |\hat{g}(b) - 0.95|$
- The grid and K are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- A non-studentized bootstrap with a fixed 6-month block is not this test: its block is not calibrated for coverage

Back

The Deflated Sharpe Ratio

- Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- Expected maximum under H_0: all true Sharpe ratios zero, N independent trials, monthly units as T; on this slide N counts trials (configurations), not assets
  - $\gamma_E = 0.5772$: Euler's constant; $\hat{V}$: the variance of the N estimates
  - Φ: standard Normal distribution function; e: Euler's number

$$SR_0 = \sqrt{\hat{V}}\left[(1-\gamma_E)\,\Phi^{-1}\left(1-\tfrac{1}{N}\right) + \gamma_E\,\Phi^{-1}\left(1-\tfrac{1}{Ne}\right)\right]$$

- Deflated Sharpe ratio: the probability that the true Sharpe of the winner exceeds SR_0, with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$\text{DSR} = \Phi\left(\frac{(\widehat{SR} - SR_0)\sqrt{T-1}}{\sqrt{1-\hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right)$$

- This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
  - As 24 independent trials: $SR_0 = 0.52$ a year, DSR $= 0.84 < 0.95$
  - The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98
- Holm adjusts the whole family of pairwise tests against 1/N; the DSR asks whether the single best configuration is better than luck

Back

Beyond Equities, Digital Assets

- Petukhina, Trimborn, Härdle & Elendner (2021): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- Pele et al. (2023): are cryptos becoming alternative assets?
- Bitcoin: annualised volatility 172% (monthly, 2010–2026, one surviving coin chosen ex post), IEF 6.6%
- Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically (Seminar 4, B8; the 172% is the check value for its first task)
- Robust mean–variance for crypto-assets: Pele (2023)

Back

Case Study, Method in Detail (1/2)

- **Cost** (eqs. 4, 7)
  - π_t: risky-asset weights; $1 - 1^\top \pi_t$: the weight of the risk-free asset
  - $g_t = \mathrm{diag}\big((1 + r_t^f + r_t)/(1 + g_t^w)\big)$ (eq. 6); r_t^f: risk-free rate, r_t: asset returns, g_t^w: growth of wealth
  - $\mathbf{\Lambda} = \mathrm{diag}(0.2/V_i)$; V_i: average daily dollar volume over 6 months; trading 1% of V_i moves the price by 0.1% (eq. 35, Frazzini, Israel & Moskowitz (2018))
- **Optimal rule** (Proposition 3): $\pi_t = m\, g_t \pi_{t-\mathbf{1}} + (\mathrm{I} - m) A_t$, generalising Gârleanu & Pedersen (2013)
  - A_t: a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14)
- **Portfolio-ML** (Proposition 4, eq. 40): $A_t = \mathrm{diag}(1/\sigma_{i,t})\, f(s_t)\beta$
  - $\sigma_{i,t}$: volatility of asset i; f: p random Fourier features of the signals s_t
  - $\hat{\beta} = (E_T[\tilde{\mathbf{\Sigma}}_t] + \lambda \mathrm{I})^{-\mathbf{1}} E_T[\tilde{r}_{t+\mathbf{1}}]$: a ridge regression ($\lambda$: penalty) that maximises utility net of costs in closed form (eqs. 24–26)
  - it learns the weights directly, as Brandt, Santa-Clara & Valkanov (2009) do without costs

Back

Case Study, Method in Detail (2/2)

- **Data** (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks)
  - 115 characteristics of Jensen, Kelly & Pedersen (2023), ranked into $[0, 1]$
  - December 1952 – December 2020, out of sample 1981–2020, monthly
- **Covariance Σ**: 13 characteristic themes and 12 industries as factors
  - exponential weights with half-lives of 378 days (correlations) and 126 days (variances)
  - investor $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market (Section 5.1.3)
- **Tuning** (Table 1), chosen each year by the utility of validation backtests since 1971
  - $\lambda \in \{0, e^{-\mathbf{10}}, e^{-\mathbf{9.8}}, \ldots, e^{\mathbf{10}}\}$, $p \in \{2^{\mathbf{6}}, \ldots, 2^{\mathbf{9}}\}$, $\eta \in \{e^{-\mathbf{3}}, e^{-\mathbf{2}}\}$; η: standard deviation of the random weights
- **Rivals** ("*" marks an extra tuning layer, Section 5.3)
  - Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22)
  - factor sorts, 1/N, minimum variance, market

Back

Case Study, Complete Results

- **Table 2**, Portfolio-ML: turnover 0.25 a month (net Sharpe ratio 1.33)
- **Table 2**, Multiperiod-ML*: net Sharpe 0.83, utility 0.020
  - Portfolio-ML needs no second tuning layer
- **Table 3**: posterior probability that Portfolio-ML has the higher utility
  - 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- **Figure 1(B)** (Section 6.2), $\gamma = 10$: size is a cost
  - with \$10 billion, net excess return 14.6% at 11% volatility; with \$1 billion, 16.3%
- **Figure 5**: net of costs, quality is the most important theme; short-term reversal is almost irrelevant
  - monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3)
  - persistent signals are cheap to trade
- **Limits stated by the authors**
  - fees and other frictions; computing power and ML methods not available in real time early in the sample (Section 6.1)
  - costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3)
  - among the smallest 20% of stocks, Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)

Back

Case Study, the Course Replication

- **Course data**: 14 US stocks and 9 sector ETFs
  - five signals: 1-month log return, log momentum from $t-6$ and from $t-12$ to $t-1$, 1-year volatility, 6-month dollar volume (formulas in Seminar 4, C3)
  - same cost rule, γ, wealth and grid; $\boldsymbol{\Sigma}$ with the same weights on the 23 assets
  - validation from 2011; out of sample February 2014 – August 2026 (151 months)
- **Portfolio-ML**: cost 0.6% a year, leverage 0.32
- **Other rules**: utility of Static-ML -0.13; minimum variance -0.72; Markowitz-ML extremely negative (leverage 18)
- **Posterior probability** of a higher utility for Portfolio-ML
  - 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- Same ranking of cost-aware against cost-agnostic rules as Table 2
  - Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- **Limits**
  - 23 assets instead of 184–805; five signals instead of 115
  - no factor model for $\boldsymbol{\Sigma}$, whose window reaches 10 years only in August 2014 (common prices from August 2004)
  - return forecasts with $p \geq 2^6$ and three validation years (paper: 2^1–2^9, ten years); no starred rivals
- **Open extension** (Seminar 4, C3)
  - implementable frontiers for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$
  - 1/N's trading costs computed at each wealth level

Back

The Discovery Loop, Reusable Prompts

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**: *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with 1/N out of sample; for each give DOI, N, T, cost assumption and test."* Resolve every DOI on Crossref and open the table cited
- **Hypotheses**: *"Give three competing explanations for a Sharpe gain over 1/N (better Σ, a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code**: *"Write Python that reads the course's daily price files, aligns prices on common dates and runs a 60-month rolling GMV-LW and 1/N; weights at t may use data up to $t-1$ only."* Replicate first: 1/N 0.76, GMV-LW 0.20 on the combined universe
- **Robustness**: *"Draft a pre-registration: the family of tests (rules $\times$ universes $\times$ windows $\times$ costs), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
- **Critique**: *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report**: every specification tried, adjusted p-values, the power of the design, code and seeds; any specification chosen after seeing results is exploratory

Back

Answers to the Quick Checks

- **Three rules**: $s_{11} = 0.09$, $s_{22} = 0.0225$, $s_{12} = 0.2 \times 0.30 \times 0.15 = 0.009$
  - GMV: $w_1 = (0.0225 - 0.009)/(0.09 + 0.0225 - 0.018) = 1/7$, volatility 14.3%; ERC (inverse volatility): $w_1 = 1/3$, volatility 15.5%; 1/N: 18.1%
  - 1/N: $w_1(\Sigma w)_1 = 0.5 \times 0.0495 = 0.02475$ of $\sigma_p^2 = 0.0326$: asset 1 carries 76% of the risk with 50% of the weight
- **Plug-in Sharpe**: monthly $\theta^2 = 0.25/12 = 0.0208$; $(120 \times 0.0208 + 4)/(120 - 6) = 0.0570$; $\sqrt{12 \times 0.0570} = 0.83$ a year, against a true 0.50
- **Noise band**: $c = 10/40 = 0.25$, band $[(1 - 0.5)^2, (1 + 0.5)^2] = [0.25, 2.25]$; 2.0 lies inside, so this comparison cannot separate it from noise
- **One view**: equal precisions give view weight $0.0009/(0.0009 + 0.0009) = 0.5$, posterior 7%; with $\Omega = 0.0003$, weight $0.0009/0.0012 = 0.75$, posterior $4\% + 0.75 \times 6\% = 8.5\%$
- **One Sharpe ratio**: SE $= \sqrt{(1 + 0.15^2/2)/120} = 0.092$; annual $0.52 \pm 1.96 \times 0.32$, interval $[-0.10, 1.14]$: not significant
- **Four tests**: Holm $0.016, 0.060, 0.060, 0.300$: one rejection; Benjamini–Hochberg $0.016, 0.040, 0.040, 0.300$: three rejections

Back

Courses, Code and Data

Advanced courses on quantinar.com

- Hierarchical Clustering
- Minimum Spanning Tree
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- Risk-free rate: Kenneth French Data Library

Back

References (1/6)

- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- Andrews, D.W.K., Monahan, J.C. (1992). An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. *Econometrica*, 60(4), 953–966.
- Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- Best, M.J., Grauer, R.R. (1991). On the sensitivity of mean-variance-efficient portfolios to changes in asset means: some analytical and computational results. *Review of Financial Studies*, 4(2), 315–342.
- Black, F., Litterman, R. (1992). Global portfolio optimization. *Financial Analysts Journal*, 48(5), 28–43.
- Bodnar, T., Parolya, N., Schmid, W. (2018). Estimation of the global minimum variance portfolio in high dimensions. *European Journal of Operational Research*, 266(1), 371–390.
- Brandt, M.W., Santa-Clara, P., Valkanov, R. (2009). Parametric portfolio policies: exploiting characteristics in the cross-section of equity returns. *Review of Financial Studies*, 22(9), 3411–3447.
- Britten-Jones, M. (1999). The sampling error in estimates of mean-variance efficient portfolio weights. *Journal of Finance*, 54(2), 655–671.
- Chopra, V.K., Ziemba, W.T. (1993). The effect of errors in means, variances, and covariances on optimal portfolio choice. *Journal of Portfolio Management*, 19(2), 6–11.

References (2/6)

- Choueifaty, Y., Coignard, Y. (2008). Toward maximum diversification. *Journal of Portfolio Management*, 35(1), 40–51.
- DeMiguel, V., Garlappi, L., Nogales, F.J., Uppal, R. (2009). A generalized approach to portfolio optimization: improving performance by constraining portfolio norms. *Management Science*, 55(5), 798–812.
- DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- Dragotă, M., Şerbănescu, C.I., Pele, D.T. (2008). Portfolio diversification and market share analysis for Romanian insurance companies. In *Risk Management and Value*, World Scientific, 277–298.
- Fan, J., Liao, Y., Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B*, 75(4), 603–680.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Frazzini, A., Israel, R., Moskowitz, T.J. (2018). Trading costs. SSRN Working Paper 3229719.
- Frost, P.A., Savarino, J.E. (1988). For better performance: constrain portfolio weights. *Journal of Portfolio Management*, 15(1), 29–34.
- Gârleanu, N., Pedersen, L.H. (2013). Dynamic trading with predictable returns and transaction costs. *Journal of Finance*, 68(6), 2309–2340.
- Härdle, W.K., Simar, L. (2019). Applied Multivariate Statistical Analysis, 5th ed. Springer.

References (3/6)

- Harvey, C.R., Liu, Y. (2015). Backtesting. *Journal of Portfolio Management*, 42(1), 13–28.
- He, G., Litterman, R. (2002). The intuition behind Black-Litterman model portfolios. SSRN Working Paper 334304.
- Jagannathan, R., Ma, T. (2003). Risk reduction in large portfolios: why imposing the wrong constraints helps. *Journal of Finance*, 58(4), 1651–1683.
- Jensen, T.I., Kelly, B., Malamud, S., Pedersen, L.H. (2026). Machine learning and the implementable efficient frontier. *Review of Financial Studies*, 39(10), 3035–3078.
- Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance?. *Journal of Finance*, 78(5), 2465–2518.
- Jobson, J.D., Korkie, B.M. (1981). Performance hypothesis testing with the Sharpe and Treynor measures. *Journal of Finance*, 36(4), 889–908.
- Jorion, P. (1986). Bayes-Stein estimation for portfolio analysis. *Journal of Financial and Quantitative Analysis*, 21(3), 279–292.
- Kan, R., Wang, X., Zhou, G. (2022). Optimal portfolio choice with estimation risk: no risk-free asset case. *Management Science*, 68(3), 2047–2068.
- Kan, R., Zhou, G. (2007). Optimal portfolio choice with parameter uncertainty. *Journal of Financial and Quantitative Analysis*, 42(3), 621–656.

References (4/6)

- Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.
- Laloux, L., Cizeau, P., Bouchaud, J.-P., Potters, M. (1999). Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7), 1467–1470.
- Ledoit, O., Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5), 603–621.
- Ledoit, O., Wolf, M. (2004a). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- Ledoit, O., Wolf, M. (2004b). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Ledoit, O., Wolf, M. (2017). Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets Goldilocks. *Review of Financial Studies*, 30(12), 4349–4388.
- Ledoit, O., Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5), 3043–3065.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.

References (5/6)

- López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management*, 42(4), 59–69.
- Maillard, S., Roncalli, T., Teïletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70.
- Mantegna, R.N. (1999). Hierarchical structure in financial markets. *European Physical Journal B*, 11(1), 193–197.
- Marčenko, V.A., Pastur, L.A. (1967). Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4), 457–483.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- Merton, R.C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4), 1851–1872.
- Merton, R.C. (1980). On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics*, 8(4), 323–361.
- Michaud, R.O. (1989). The Markowitz optimization enigma: is 'optimized' optimal?. *Financial Analysts Journal*, 45(1), 31–42.
- Okhrin, Y., Schmid, W. (2006). Distributional properties of portfolio weights. *Journal of Econometrics*, 134(1), 235–256.

References (6/6)

- Pele, D.T. (2023). Robust Markowitz for crypto-assets. Quantinar course, Bucharest University of Economic Studies.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- Petukhina, A., Trimborn, S., Härdle, W.K., Elendner, H. (2021). Investing with cryptocurrencies – evaluating their potential for portfolio allocation strategies. *Quantitative Finance*, 21(11), 1825–1853.
- Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Sharpe, W.F. (1966). Mutual fund performance. *Journal of Business*, 39(S1), 119–138.
- Theil, H., Goldberger, A.S. (1961). On pure and mixed statistical estimation in economics. *International Economic Review*, 2(1), 65–78.
- Tobin, J. (1958). Liquidity preference as behavior towards risk. *Review of Economic Studies*, 25(2), 65–86.
- Tu, J., Zhou, G. (2011). Markowitz meets Talmud: a combination of sophisticated and naive diversification strategies. *Journal of Financial Economics*, 99(1), 204–215.
- Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Acronyms used in this material

- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **DSR**: Deflated Sharpe Ratio
- **ERC**: Equal Risk Contribution portfolio
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMV**: Global Minimum-Variance portfolio
- **GMV-LO**: Global Minimum-Variance portfolio, Long-Only
- **GMV-LW**: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HRP**: Hierarchical Risk Parity
- **HYG**: iShares iBoxx High Yield Corporate Bond ETF
- **IEF**: iShares 7–10 Year Treasury Bond ETF
- **LO**: Long-Only, no short sales
- **LQD**: iShares iBoxx Investment Grade Corporate Bond ETF
- **ML**: Machine Learning
- **MV**: Mean–Variance tangency portfolio estimated from sample moments
- **MV-LO**: Mean–Variance tangency portfolio, Long-Only
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **OOS**: Out-Of-Sample
- **POET**: Principal Orthogonal complEment Thresholding
- **US**: United States